

Mobile Applications Development

Recipe Finder Ionic App

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Introduction

Write an Ionic mobile application that:

- reads recipe data from the internet (Spoonacular.com)
- allows a search for recipes by ingredients
- allows viewing of full recipe details (ingredients, instructions)
- maintains a favourite recipes list
- allows measurement units to be shown in either US or Metric.

Project Start

The project must be written using the Ionic framework V7.2.1 (the version on the Virtual Machine (VM)) and created initially using the following command:

```
ionic start GXXXXXXXX blank --type=angular-standalone
```

where GXXXXXXXX is your student number.

The project will be tested on the VM, and must work with the default software that was installed on that machine.

Submission of the Project

You only need to zip the *src* folder (the entire project will be too big).

It will be tested by copying the *src* folder into a previously created project as described above.

The zipped file should be named GXXXXXXXX.zip (where GXXXXXXXX is your student number) and should be uploaded to the *Project* section of Moodle in the Mobile Applications Development module no later than **Monday, January 5th 2026 at 9:00am.**

Your project should be developed over time in GitHub, with each commit reflecting the changes you have made.

It is not sufficient to simply have one or two commits.

A file entitled *GitLink.pdf* (or *GitLink.txt*) must be included in the *src* folder of your application.

Ensure your repository is publicly accessible.

Plagiarism

Plagiarism will be dealt with in accordance with the university's [Student Code](#).

Marks

Marking Scheme

80% of the marks will be awarded for implementing the functionality described in this document.

20% of the marks will be awarded for innovation, extra functionality, exceeding the requirements listed in this document.

Please describe your innovation (if any) in a document entitled *innovation.pdf* which should be stored in the *src* folder of your submission.

Viva

You may be asked to explain your project in full, or in part, during an online [viva](#).

Overview of App

The app should consist of the following pages:

Home Page

The *Home Page* is the first page of the application and should show your student number, a “Favourites” icon, and a “Settings” icon.

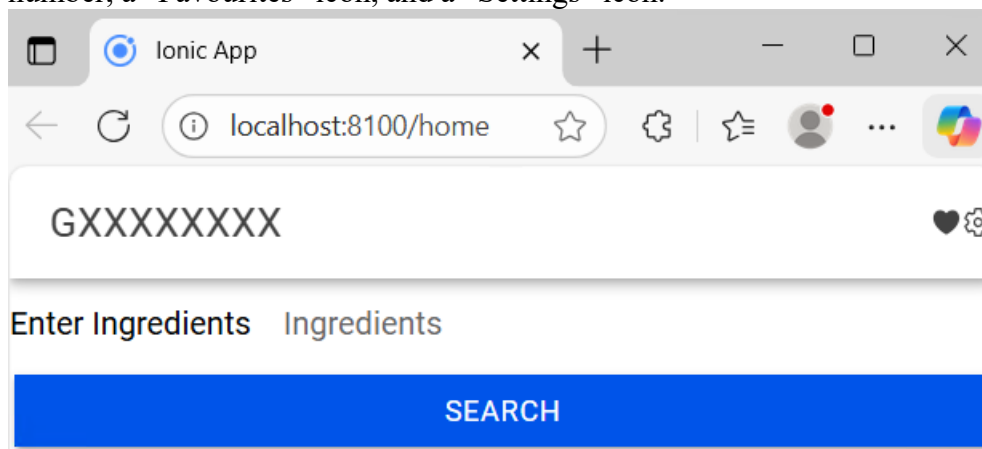


Figure 1 Home Page

The *Home Page* should allow the user to enter one ingredient, or a number of ingredients, to find recipes based on. Multiple ingredients should be separated by commas.

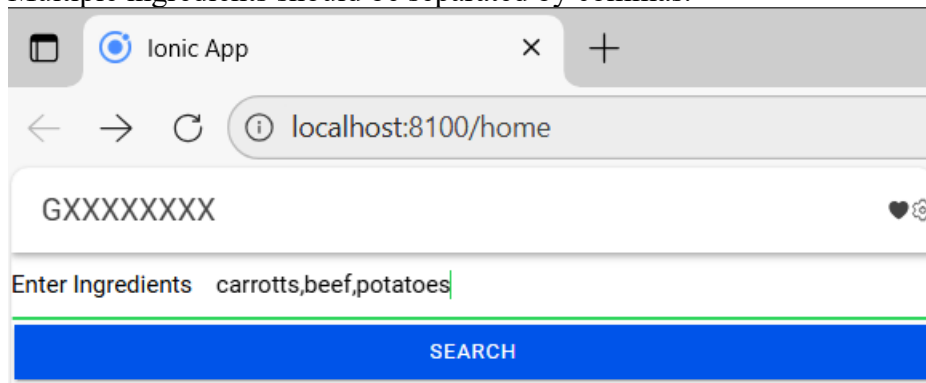


Figure 2 Home Page with text entered

When the *Search* button is pressed any recipes (from Spoonacular.com) that match the ingredients should be displayed.

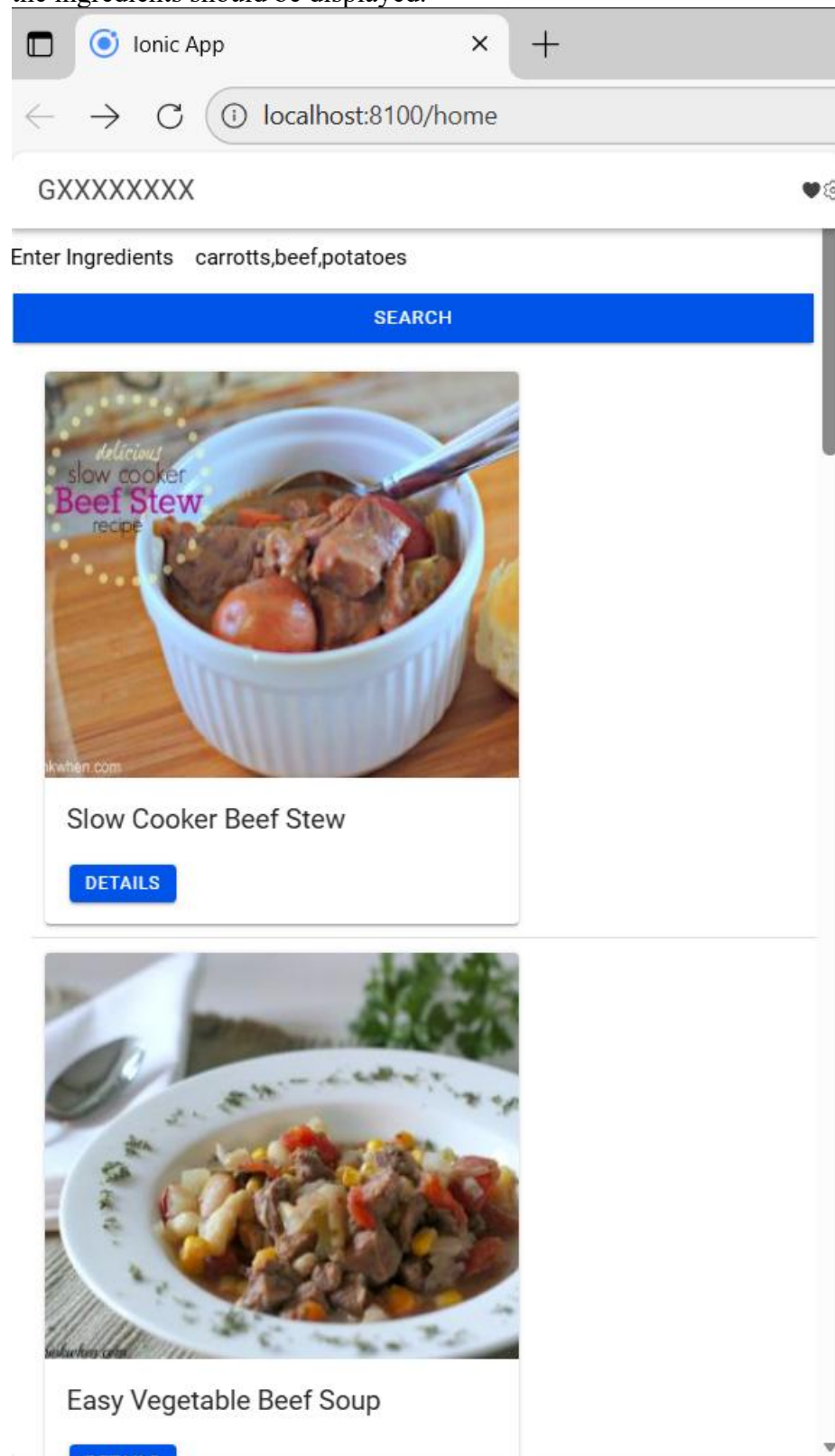


Figure 3 Recipes based on ingredients

Recipe Details Page

When the user presses *Details* on any recipe, the full recipe details should be shown on another page.

If the user selected *Metric* on the *Settings* page, the units should be shown in Metric.

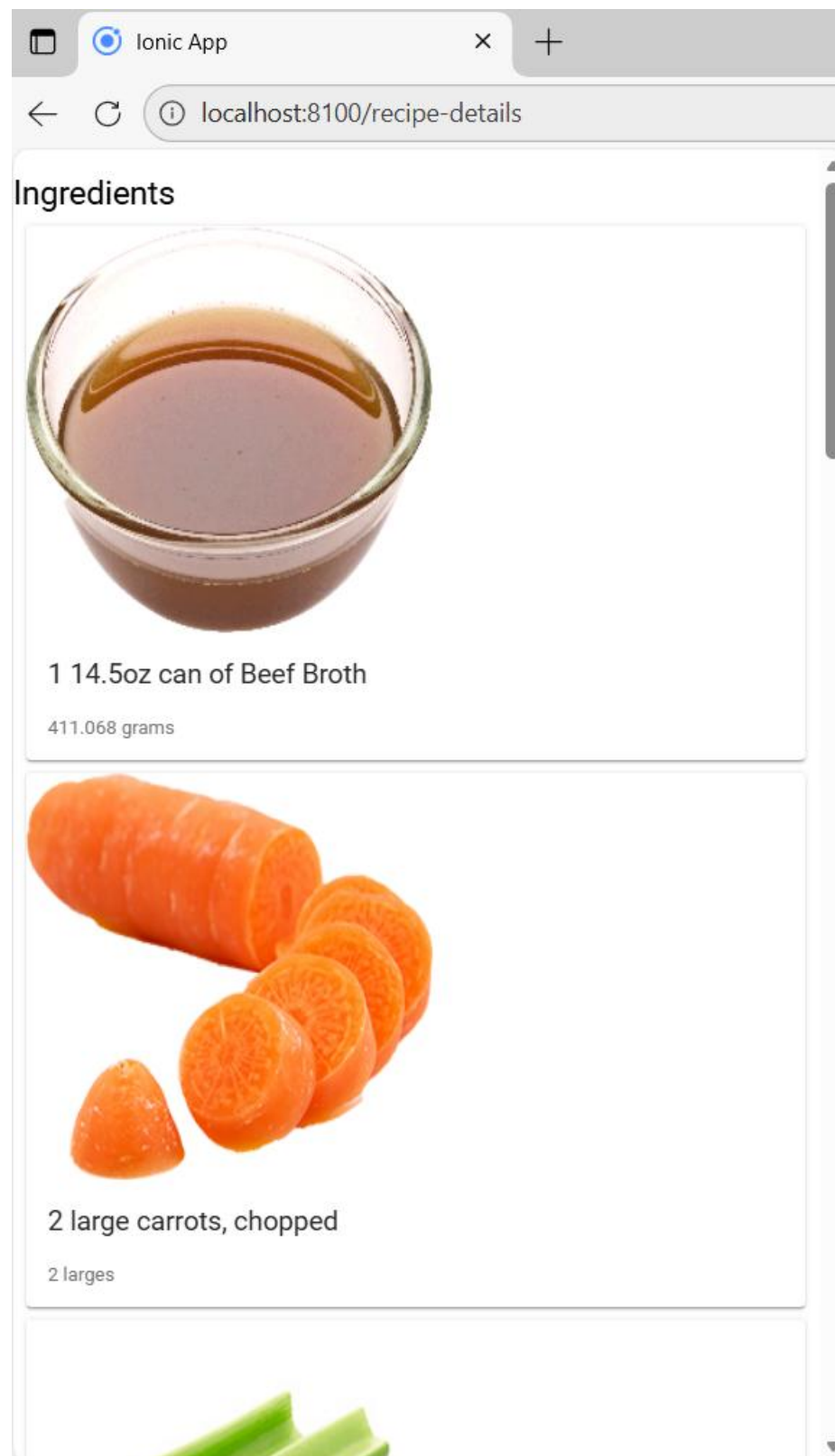


Figure 4 Ingredients of selected recipe (units in Metric)

If the user selected *US* on the *Settings* page, the units should be shown in US measurements.

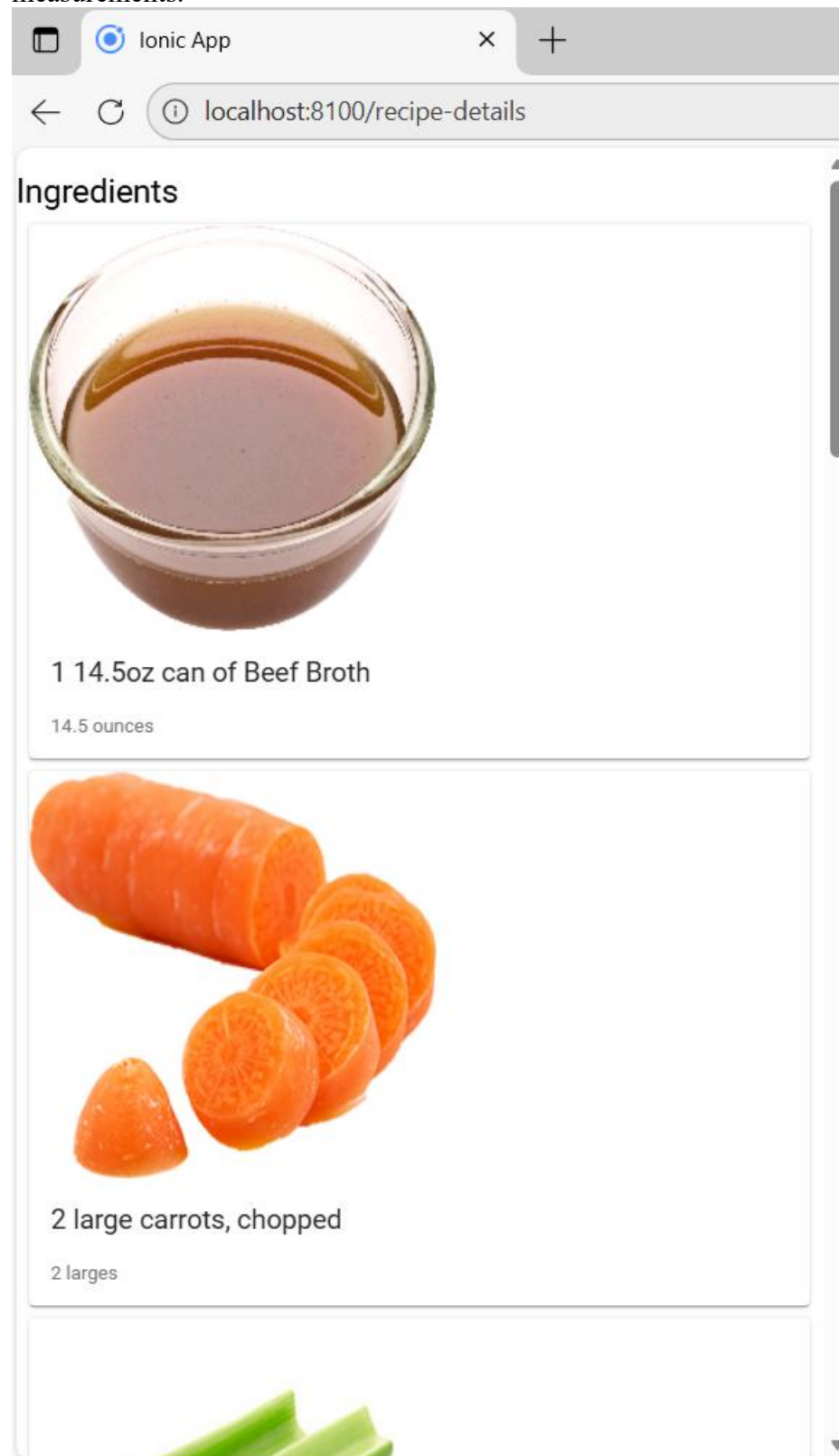


Figure 5 Ingredients of selected recipe (units in US)

After all the ingredients have been listed the instructions should be shown, as well as a button entitled “*Add to Fvaourites*”.

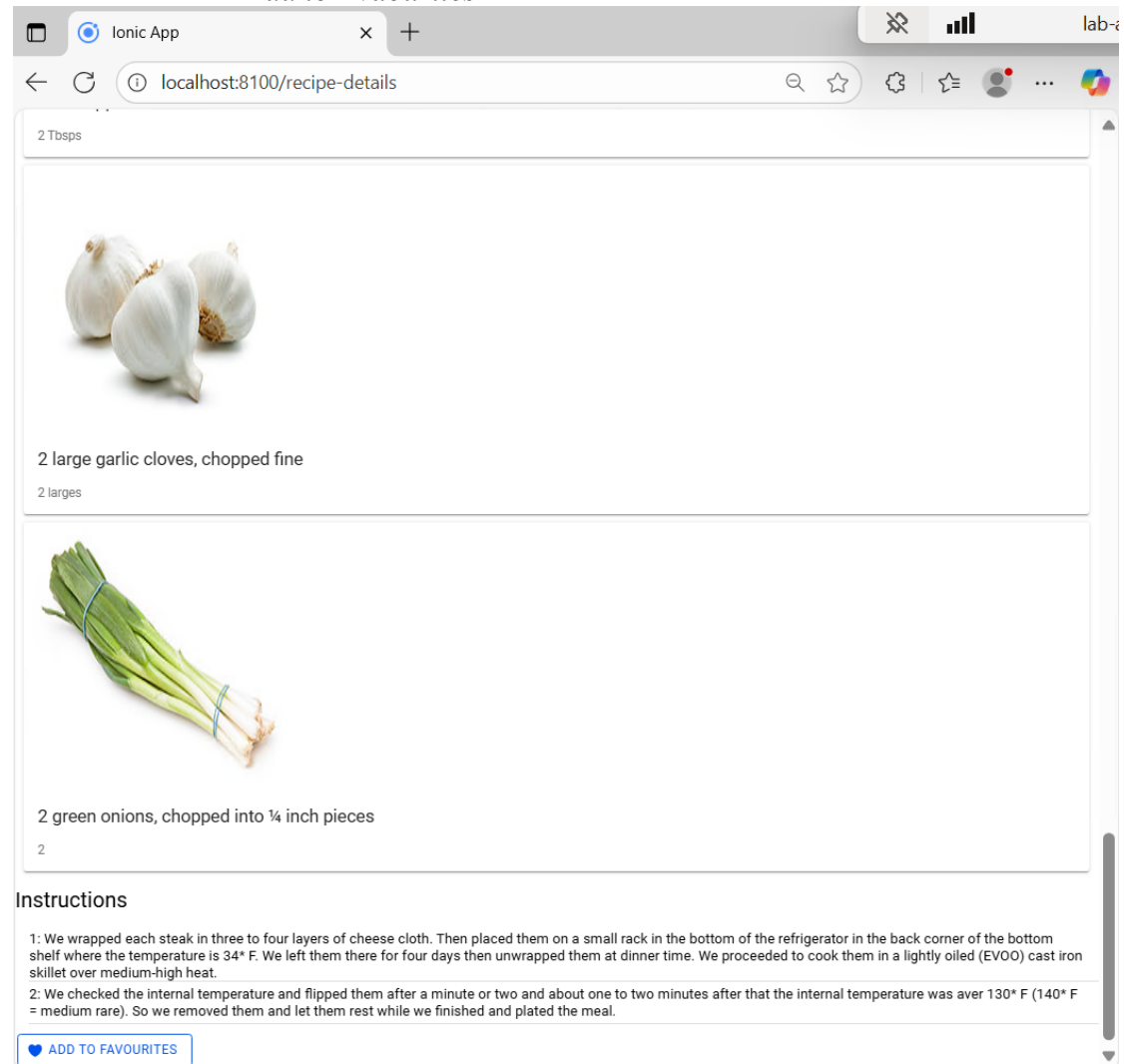


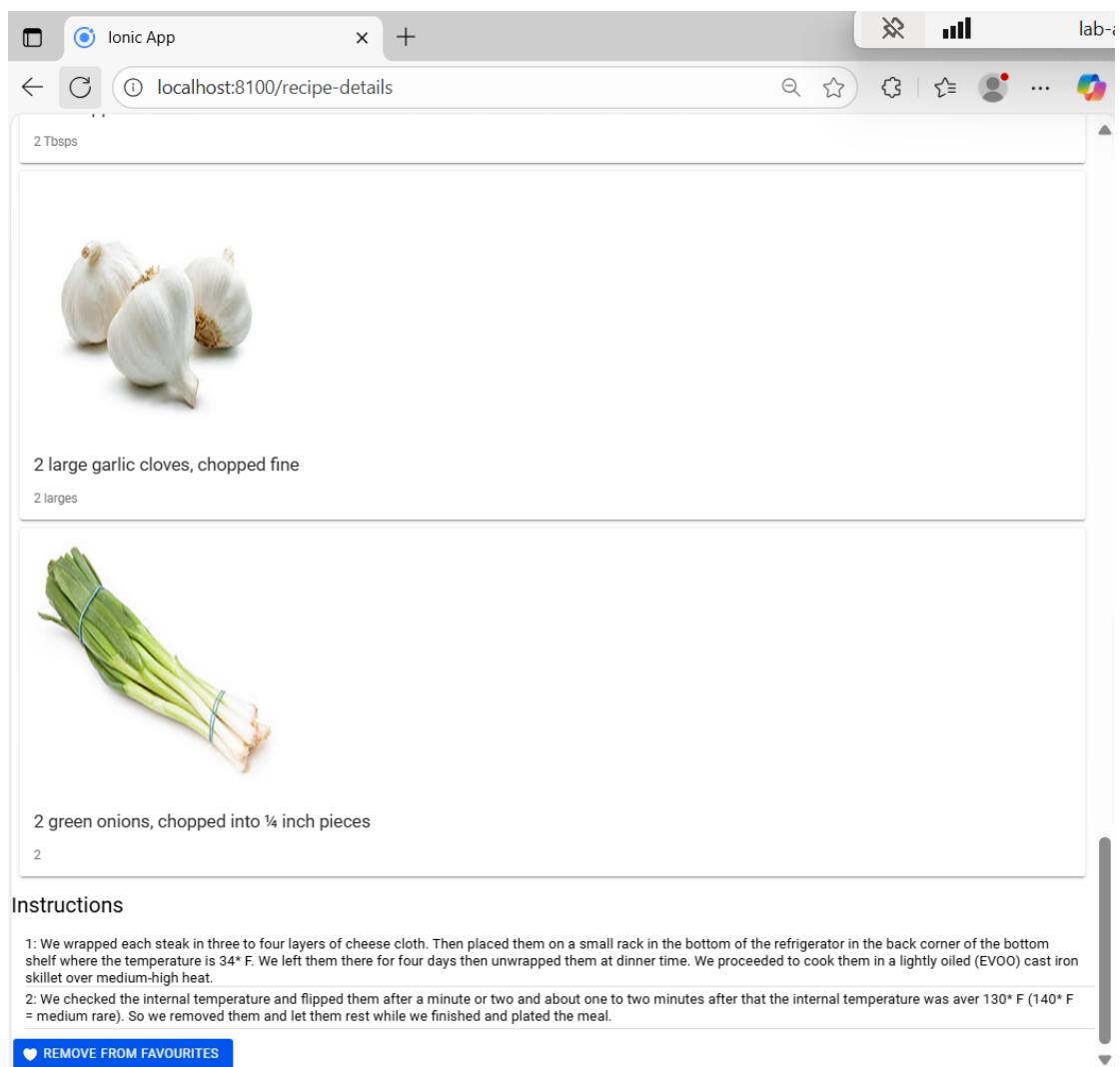
Figure 6 Recipe Details Page – Instructions

Add To Favourites Button

If the selected recipe is not already a “Favourite” the button should be entitled “*Add to Favourites*”. When pressed it should add this recipe to a “Favourites” list, and the button title should change to “*Remove From Favourites*”.

If the selected recipe is already a “Favourite”, the button should be entitled “*Remove From Favourites*”. When pressed it should remove this recipe from the “Favourites” list, and the button title should change to “*Add to Favourites*”.

Any additions to, or removals from, the “Favourites List” should be persisted after the application has ended.



Settings Page

When the “settings” icon is pressed the *Settings Page* should be opened. It should display the following list, with the previously selected “Measurement” selected. If no “Measurement” has been previously selected (e.g. this is the first time the app has been run, or the database has been deleted) the value “Metric” should be selected by default.

Any changes made here should be persisted after the application has ended.

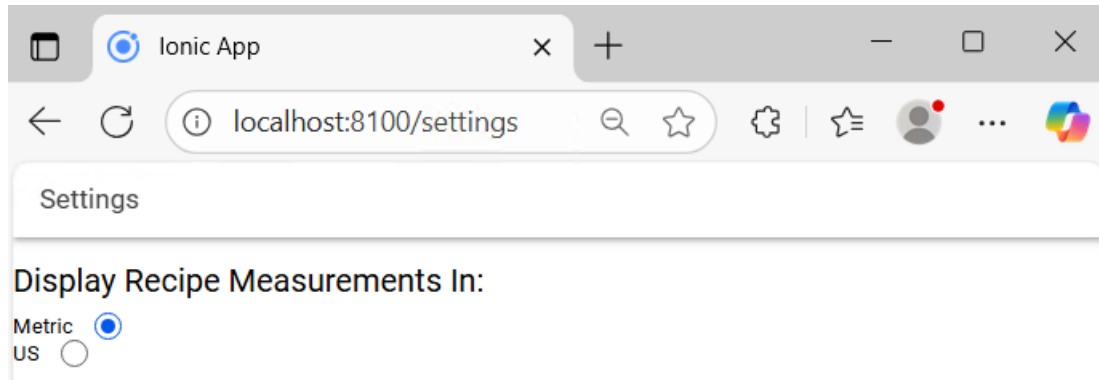


Figure 7 Settings Page

Favourites Page

When the “favourites/heart” icon is pressed the *Favourites Page* should be opened. It should display any recipe the user has pressed the “*Add To Favourites*” button on.

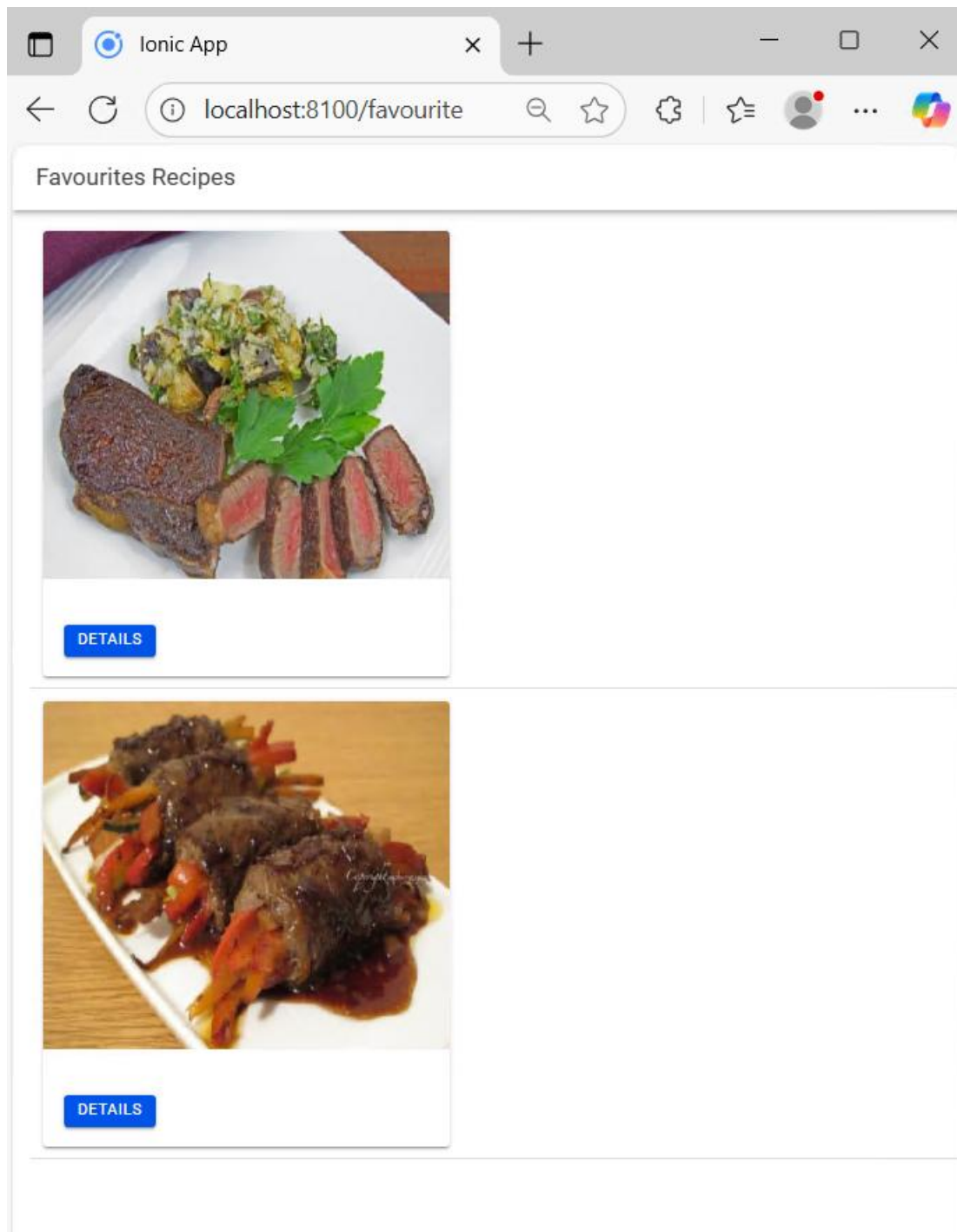


Figure 8 Favourites Page

Details Button

When the *Details* button is pressed, Recipe Details Page should be open for that recipe.

API Details

The API (Application Programming Interface) used will be spoonacular.com.

The API key you should use is: [70759a4f7911402abcc53d3c51d3b759](#).
(You don't need to sign up to the API).

API used on Home Page

To find all recipes containing the ingredient(s) entered on the search page use:
<https://spoonacular.com/food-api/docs#Search-Recipes-Complex>

The only request parameters of interest are:

- **query**
Which is the ingredient(s) entered on the Home Page. If more than one, they should be separated by commas.
- **apiKey**
Which is [70759a4f7911402abcc53d3c51d3b759](#).

The response parameters of interest are:

- **id**
Which is the recipe ID, which will be used if the user presses the “Details” button.
- **title**
Which is the text that should be displayed.
- **image**
Which is the image that should be displayed.

The following image shows the relationship between what is returned from the API, and what is displayed:

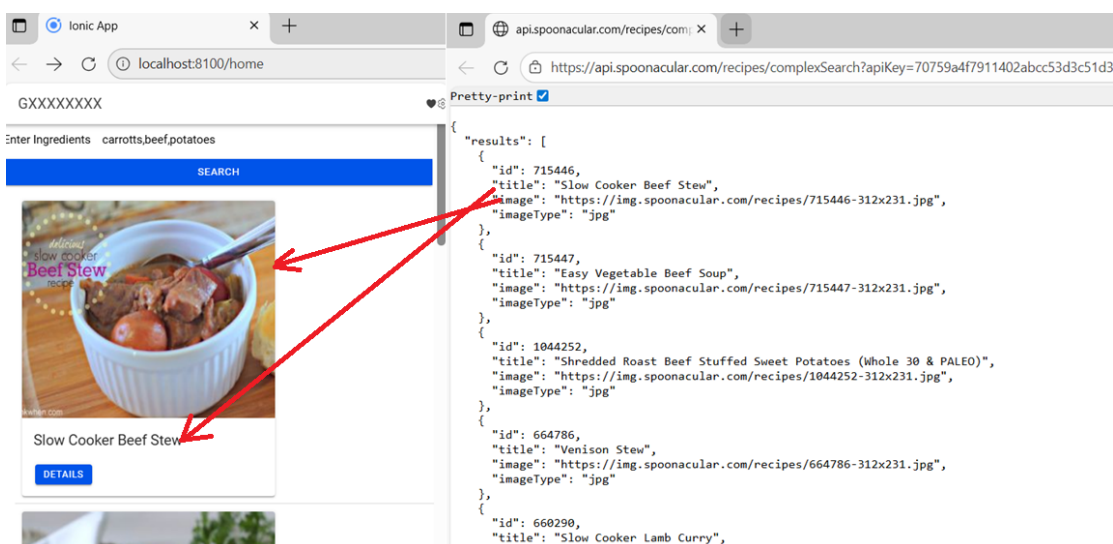


Figure 9 What is used from the API

API used on Recipe Details Page.

When the “*Details*” button is pressed for a particular recipe, the Recipe Details Page is open showing more information on the selected recipe using the following API endpoint:

<https://spoonacular.com/food-api/docs#Get-Recipe-Information>

The only request parameters of interest are:

- **id**
Which is ID of the recipe on which the “*Details*” button was pressed.
- **apiKey**
Which is **70759a4f7911402abcc53d3c51d3b759**.

The response parameters of interest are:

- **image**
Which is the image of the ingredient.
- **extendedIngredients**
 - **original**
Which is the name of the ingredient.
- **measures**
 - **us**
 - **amount**
Which is the amount of the ingredient to use (if *US* is selected in the *Settings* page).
 - **unitLong**
Which is the units of the ingredient to use (if *US* is selected in the *Settings* page).
 - **metric**
 - **amount**
Which is the amount of the ingredient to use (if *Metric* is selected in the *Settings* page).
 - **unitLong**
Which is the amount of the ingredient to use (if *Metric* is selected in the *Settings* page).
- **analyzedInstructions**
 - **steps**
 - **number**
Which is the step number.
 - **step**
Which is the step instruction.

The following image shows the relationship between what is returned from the API, and what is displayed if the user selected *US* on the *Settings* page:

The screenshot shows a web application interface for recipe details. On the left, under the heading "Ingredients", there are two items: a bowl of brown liquid (beef broth) and a pile of orange carrot slices. Below the broth image is the text "1 14.5oz can of Beef Broth" and "14.5 ounces". Below the carrot image is the text "2 large carrots, chopped" and "2 larges". On the right, a JSON object represents the recipe data. Red arrows point from the UI elements to the JSON fields: from the broth image to the "image" field of the first ingredient entry, from the text "1 14.5oz can of Beef Broth" to the "originalName" field, from the text "14.5 ounces" to the "amount" field, and from the text "2 large carrots, chopped" to the "amount" field of the second ingredient entry.

```
{
  "id": 715446,
  "image": "https://img.spoonacular.com/recipes/715446-556x370.jpg",
  "imageType": "jpg",
  "title": "Slow Cooker Beef Stew",
  "readyInMinutes": 490,
  "servings": 6,
  "sourceUrl": "https://www.pinkwhen.com/slow-cooker-beef-stew-recipe/",
  "vegetarian": false,
  "vegan": false,
  "glutenFree": true,
  "dairyFree": true,
  "veryHealthy": true,
  "cheap": false,
  "veryPopular": false,
  "sustainable": false,
  "lowFodmap": false,
  "weightWatcherSmartPoints": 10,
  "gaps": "no",
  "preparationMinutes": 10,
  "cookingMinutes": 480,
  "aggregateLikes": 57,
  "healthScore": 100,
  "creditsText": "pinkwhen.com",
  "license": null,
  "sourceName": "pinkwhen.com",
  "pricePerServing": 270.41,
  "extendedIngredients": [
    {
      "id": 6008,
      "aisle": "Canned and Jarred",
      "image": "beef-broth.png",
      "consistency": "LIQUID",
      "name": "beef broth",
      "nameClean": "beef broth",
      "original": "1 14.5oz can of Beef Broth",
      "originalName": "Beef Broth",
      "amount": 14.5,
      "unit": "oz",
      "meta": [
        "canned"
      ],
      "measures": {
        "us": {
          "amount": 14.5,
          "unitShort": "oz",
          "unitLong": "ounces"
        },
        "metric": {
          "amount": 411.068,
          "unitShort": "g",
          "unitLong": "grams"
        }
      }
    },
    {
      "id": 10156,
      "aisle": "Produce",
      "image": "carrots.png",
      "consistency": "SOLID",
      "name": "carrots",
      "nameClean": "carrots",
      "original": "2 large carrots, chopped",
      "originalName": "Carrots",
      "amount": 2,
      "unit": "large",
      "meta": [
        "chopped"
      ],
      "measures": {
        "us": {
          "amount": 2,
          "unitShort": "large",
          "unitLong": "large"
        },
        "metric": {
          "amount": 132.007,
          "unitShort": "g",
          "unitLong": "grams"
        }
      }
    }
  ]
}
```

Figure 10 Ingredients in US measurements

The following image shows the relationship between what is returned from the API, and what is displayed if the user selected *Metric* on the *Settings* page:

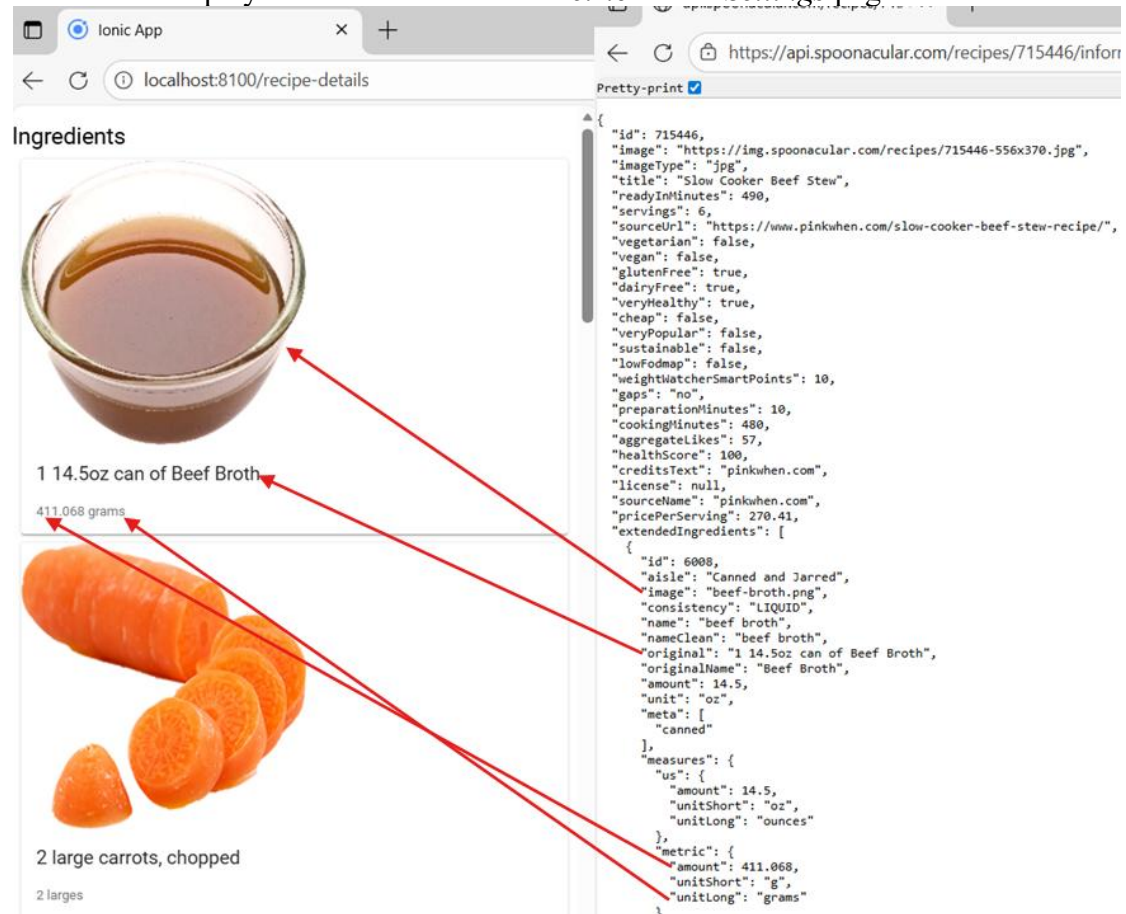


Figure 11 Ingredients in Metric measurements

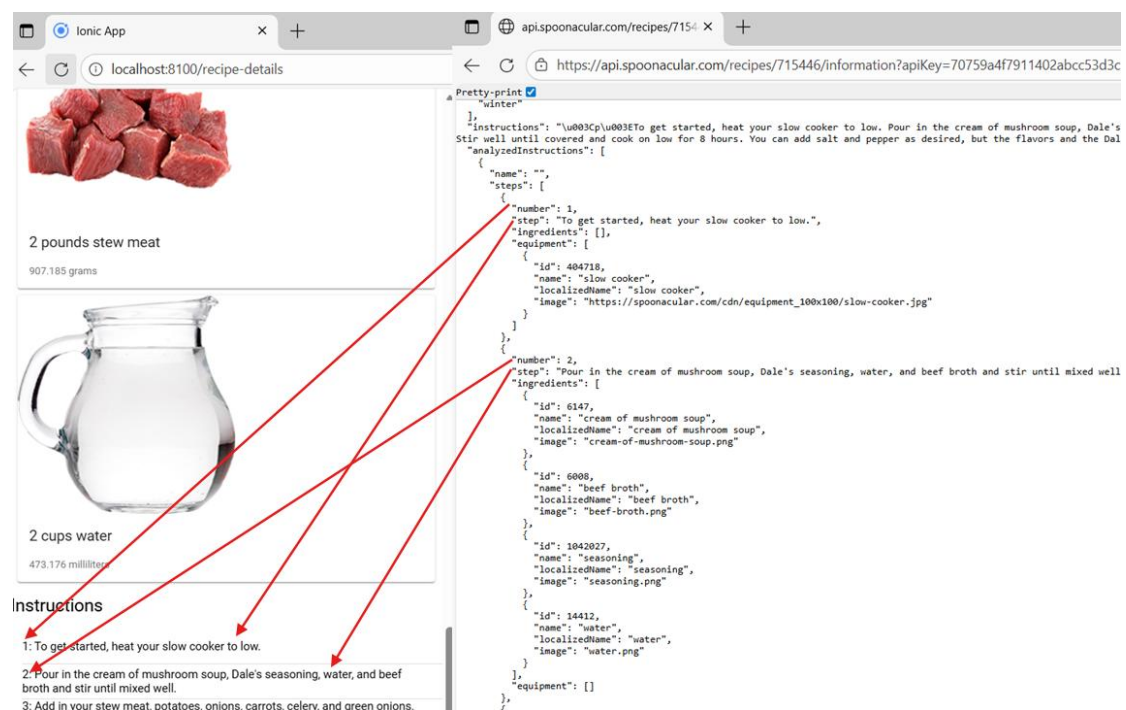


Figure 12 Instruction Steps